

COMPILER DESIGN ASSIGNMENT 1

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E SECTION

LEXER.L

```
%{
#include "parser.tab.h"
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

int line = 1;
char current_token[100];
%}

digit [0-9]
letter [_A-Za-z]

%%

"int"    { strcpy(current_token, yytext); return INT; }
"float"  { strcpy(current_token, yytext); return FLOAT; }
"char"   { strcpy(current_token, yytext); return CHAR; }
"double" { strcpy(current_token, yytext); return DOUBLE; }

"if"     { strcpy(current_token, yytext); return IF; }
"else"   { strcpy(current_token, yytext); return ELSE; }
"do"     { strcpy(current_token, yytext); return DO; }
"while"  { strcpy(current_token, yytext); return WHILE; }
"for"    { strcpy(current_token, yytext); return FOR; }
"switch" { strcpy(current_token, yytext); return SWITCH; }
"case"   { strcpy(current_token, yytext); return CASE; }
"default" { strcpy(current_token, yytext); return DEFAULT; }
"break"  { strcpy(current_token, yytext); return BREAK; }

"&&"    { strcpy(current_token, yytext); return AND; }
"||"    { strcpy(current_token, yytext); return OR; }

"++"    { strcpy(current_token, yytext); return INC; }

{letter}{letter}{digit}* {
    strcpy(current_token, yytext);
    return ID;
}
```

```

}

{digit}+ {
    strcpy(current_token, yytext);
    return NUM;
}

"=="    { strcpy(current_token, yytext); return EQ; }
"!="    { strcpy(current_token, yytext); return NE; }
"<="    { strcpy(current_token, yytext); return LE; }
">="    { strcpy(current_token, yytext); return GE; }
"<"     { strcpy(current_token, yytext); return LT; }
">"     { strcpy(current_token, yytext); return GT; }

"+"      { strcpy(current_token, yytext); return PLUS; }
"-"      { strcpy(current_token, yytext); return MINUS; }
"*"      { strcpy(current_token, yytext); return MUL; }
"/"      { strcpy(current_token, yytext); return DIV; }
"="      { strcpy(current_token, yytext); return ASSIGN; }

"("      { strcpy(current_token, yytext); return LP; }
")"      { strcpy(current_token, yytext); return RP; }
"{"      { strcpy(current_token, yytext); return LB; }
"}"      { strcpy(current_token, yytext); return RB; }
"["      { strcpy(current_token, yytext); return LSB; }
"]"      { strcpy(current_token, yytext); return RSB; }
";"      { strcpy(current_token, yytext); return SEMI; }
","      { strcpy(current_token, yytext); return COMMA; }
":"      { strcpy(current_token, yytext); return COLON; }

[ \t\r]+ ;
\n      { line++; }

. {
    strcpy(current_token, yytext);
    return yytext[0];
}

%%

int yywrap() { return 1; }

```

PARSER.Y

```
%{
```

```

#include <stdio.h>
#include <stdlib.h>
#include <string.h>

extern int line;
extern char current_token[100];

int yylex();
void yyerror(const char *s);
%}

%token INT FLOAT CHAR DOUBLE
%token IF ELSE DO WHILE FOR SWITCH CASE DEFAULT BREAK
%token ID NUM
%token PLUS MINUS MUL DIV ASSIGN INC
%token EQ NE LT GT LE GE
%token AND OR
%token LP RP LB RB LSB RSB SEMI COMMA COLON

%start program

%%

program
: declarations statements
;

/* ----- DECLARATIONS ----- */

declarations
: declarations declaration
|
;

declaration
: type declarator_list SEMI
;

type
: INT
| FLOAT
| CHAR
| DOUBLE
;

```

```
declarator_list
: declarator
| declarator_list COMMA declarator
;
```

```
declarator
: ID
| ID ASSIGN expression
| ID array_dims
| ID array_dims ASSIGN expression
;
```

```
array_dims
: array_dims LSB NUM RSB
| LSB NUM RSB
;
```

```
/* ----- STATEMENTS ----- */
```

```
statements
: statements statement
|
;
```

```
statement
: expression_stmt
| selection_stmt
| iteration_stmt
| switch_stmt
| compound_stmt
| BREAK SEMI
;
```

```
compound_stmt
: LB statements RB
;
```

```
expression_stmt
: expression SEMI
| SEMI
;
```

```
/* ----- IF ----- */
```

selection_stmt
: IF LP expression RP statement
| IF LP expression RP statement ELSE statement
;

/* ----- LOOPS ----- */

iteration_stmt
: DO statement WHILE LP expression RP SEMI
| WHILE LP expression RP statement
| FOR LP opt_expr SEMI opt_expr SEMI opt_expr RP statement
;

opt_expr
: expression
|
;

/* ----- SWITCH ----- */

switch_stmt
: SWITCH LP expression RP LB case_list RB
;

case_list
: case_list case_stmt
| case_stmt
;

case_stmt
: CASE NUM COLON statements
| DEFAULT COLON statements
;

/* ----- EXPRESSIONS ----- */

expression
: expression COMMA assignment
| assignment
;

assignment
: ID ASSIGN assignment

```
| logical  
;
```

```
logical  
: logical AND relational  
| logical OR relational  
| relational  
;
```

```
relational  
: relational EQ additive  
| relational NE additive  
| relational LT additive  
| relational GT additive  
| relational LE additive  
| relational GE additive  
| additive  
;
```

```
additive  
: additive PLUS term  
| additive MINUS term  
| term  
;
```

```
term  
: term MUL factor  
| term DIV factor  
| factor  
;
```

```
factor  
: LP expression RP  
| ID  
| ID INC  
| NUM  
;
```

```
%%
```

```
void yyerror(const char *s) {  
    printf("Syntax error at line %d, token '%s': %s\n",  
        line, current_token, s);  
}
```

```

int main() {
    if (yyparse() == 0)
        printf("Syntax valid\n");
    return 0;
}

```

RESULT:

```

[maanyabhardwaj@Maanyas-MacBook-Air ~ % cd cd_as1
[maanyabhardwaj@Maanyas-MacBook-Air cd_as1 % nano lexer.l
[maanyabhardwaj@Maanyas-MacBook-Air cd_as1 % nano parser.y

```

```

parser.y: conflicts: 3 shift/reduce
[maanyabhardwaj@Maanyas-MacBook-Air cd_as1 % nano valid.c
maanyabhardwaj@Maanyas-MacBook-Air cd_as1 % nano invalid.c

```

```

[maanyabhardwaj@Maanyas-MacBook-Air cd_as1 % ./parser < valid.c
Syntax valid
[maanyabhardwaj@Maanyas-MacBook-Air cd_as1 %
./parser < invalid.c
[Syntax error at line 3, token 'i': syntax error
maanyabhardwaj@Maanyas-MacBook-Air cd_as1 %

```

Valid test:

```

int a=5, b, c, d=10;
int x[15];
int y[10][10];
int z[1][2][3][4];

```

```

a = a + d;

```

Invalid test :

1.

```
int i;
```

```
for(i=0 i<10; i++) {  
    i++;  
}
```

2.

```
int a[];
```

```
maanyabhardwaj@Maanyas-MacBook-Air cd_as1 % nano invali.c  
maanyabhardwaj@Maanyas-MacBook-Air cd_as1 %  
[./parser < invali.c  
Syntax error at line 1, token ']': syntax error  
maanyabhardwaj@Maanyas-MacBook-Air cd_as1 %
```